Self-Concordant Reparameterization for Multivariate Gaussian Regression

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Abstract

Multivariate regression with correlated Gaussian errors is a fundamental problem in statistics and machine learning. While classical approaches rely on nonconvex joint formulations, recent advances leverage a natural-parameter reparameterization to form a jointly convex objective. However, the resulting likelihood lacks global Lipschitz-gradient continuity, rendering standard convergence analyses and fixed step-size rules for gradient and Newton-type methods inapplicable. In this paper, we establish that the reparameterized Gaussian loss is strictly self-concordant. This geometric property allows us to rigorously apply standard composite self-concordant theory, leading to a damped proximal–Newton method with explicit step sizes based on the Newton decrement. We complement this principled second-order method with a proximal–gradient baseline and evaluate the computational trade-offs through extensive simulations and a real-data application on protein expression networks.

1 Introduction

In many applications—such as mapping gene and protein regulatory networks (Friedman et al., 2008; Yin and Li, 2011; Cai et al., 2013) or econometric systems (Demirer et al., 2018; Barigozzi and Brownlees, 2019)—one observes multiple responses whose error dependence structure is scientifically meaningful. In such settings, one would like to estimate regression effects and the error precision jointly, often under structural constraints such as sparsity.

Concretely, given independent observations $\{(x_i, y_i)\}_{i=1}^n$ with predictors $x_i \in \mathbb{R}^p$ (where x_i may include an intercept term) and responses $y_i \in \mathbb{R}^q$, we consider the model $y_i \mid x_i \sim \mathcal{N}_q(\beta^\top x_i, \Sigma)$ with regression coefficients $\beta \in \mathbb{R}^{p\times q}$ and precision matrix $\Omega = \Sigma^{-1}$.

This model and related estimators have been studied extensively (see, e.g., Rothman et al., 2010; Cai et al., 2013; Yin and Li, 2011; Rai et al., 2012; Lee and Liu, 2012; Chen and Banerjee, 2017). A straightforward approach is a two-step procedure: estimate β, then estimate a sparse precision matrix from the resulting residuals, for example via the graphical lasso or CLIME (Friedman et al., 2008; Cai et al., 2011). A complementary direction is the joint estimation of (β, Ω); a classical example is the MRCE method (Rothman et al., 2010). However, joint objectives are generally nonconvex in the classical parameterization (β, Ω), which can cause local optimization methods to converge to non-global stationary points depending on initialization.

To address this nonconvexity, following Zhu (2020), the model is reparameterized via the Gaussian natural parameter $\gamma = \beta\Omega$ and optimization is carried out in (γ, Ω), under which the negative log-likelihood becomes jointly convex. This leads to the following composite convex optimization problem:

$$(\hat{\gamma}, \hat{\Omega}) \in \arg \min_{\gamma \in \mathbb{R}^{p \times q},\ \Omega \in \mathbb{S}^q_{++}} \Big\{ \widetilde{\mathcal{L}}(\gamma, \Omega) + \lambda_\gamma \|\gamma_{-1}\|_1 + \lambda_\Omega \|\Omega\|_{1,\text{off}} \Big\}, \tag{1.1}$$

where $\widetilde{\mathcal{L}}(\gamma, \Omega)$ is the reparameterized Gaussian negative log-likelihood (defined in (2.1)), γ_{-1} denotes all rows of γ except the intercept, and $\|\cdot\|_{1,\text{off}}$ is the off-diagonal ℓ_1 penalty. We refer to the resulting joint estimator as SCR-MRCE (Self-Concordant Reparameterization for MRCE). The final regression estimates are recovered via $\hat{\beta} = \hat{\gamma}\hat{\Omega}^{-1}$.

While this convexity permits standard first-order composite methods, the geometry of $\widetilde{\mathcal{L}}(\gamma, \Omega)$ is non-Euclidean: because the loss involves Ω^{-1}, $\nabla\widetilde{\mathcal{L}}$ is not globally Lipschitz in a fixed Euclidean norm. As a result, classical fixed-step analyses and step-size choices based on L-smoothness are not directly applicable.

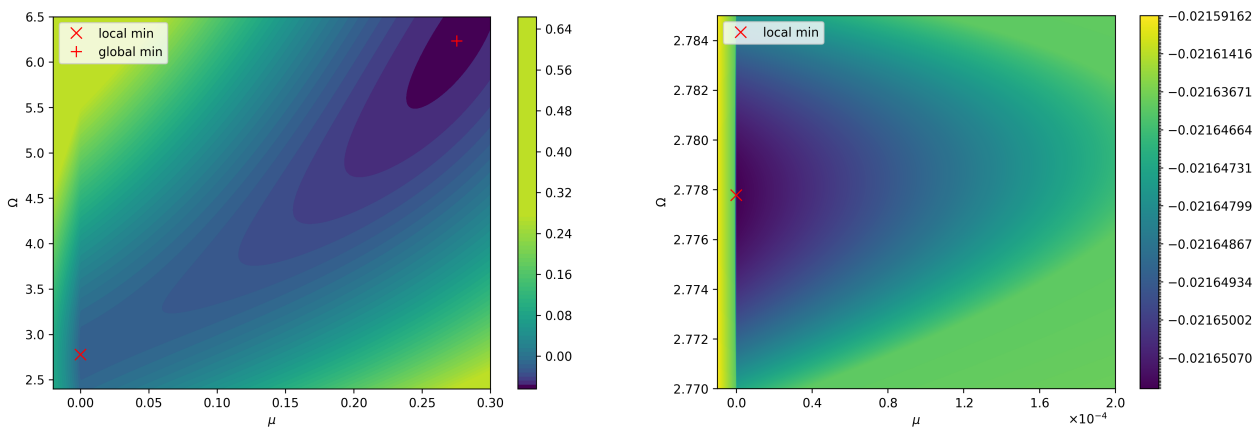
To address this geometric challenge, we exploit the objective's second-order structure. We analytically prove that the scaled loss $\mathcal{L} := 2\,\widetilde{\mathcal{L}}$ (equivalently, dropping the leading $1/2$ factor from the negative log-likelihood) is a *standard* self-concordant function. While this constant scaling is inconsequential for statistical estimation—as it is simply absorbed into the regularization parameters—it provides exact, classical curvature control for optimization. Leveraging this mathematical property, we propose a principled damped proximal–Newton method with provable global convergence and fast local convergence (via a tuning-free Newton-decrement step-size rule). We evaluate SCR-MRCE against a proximal–gradient baseline in simulations and a real-data application.

1.1 A one-dimensional motivating example

To concretely illustrate the nonconvex optimization landscape in the classical mean–precision parameterization, consider a simple one-dimensional Gaussian setting. Let $\bar{x} \in \mathbb{R}$ and $S \geq 0$ denote the sample mean and the empirical variance. For a mean parameter $\mu \in \mathbb{R}$ and a precision parameter $\Omega > 0$, consider the penalized negative log-likelihood:

$$-\log \Omega \;+\; S\,\Omega \;+\; \Omega\,(\mu - \bar{x})^2 \;+\; \lambda_1\,|\mu| \;+\; \lambda_2\,\Omega, \qquad (\mu, \Omega) \in \mathbb{R} \times (0, \infty). \tag{1.2}$$

For suitable choices of $(\bar{x}, S, \lambda_1, \lambda_2)$, this objective exhibits multiple local minima, as illustrated in Figure 1. This phenomenon persists in higher dimensions when working in (β, Ω), whereas the natural-parameter reparameterization (γ, Ω) leads to a convex global landscape.



(a) Global landscape of the objective.

(b) Zoom near a local minimum.

Figure 1: Penalized univariate Gaussian example with $\bar{x} = 0.5$, $S = 0.01$, $\lambda_1 = 2.8$, and $\lambda_2 = 0.10$. The classical (μ, Ω) parameterization yields multiple local minima.

1.2 Contributions and outline

Our main contributions are as follows:

- **Self-concordant geometry:** We show that the scaled convex Gaussian loss $\mathcal{L}$ in the (γ, Ω) parameterization is *standard* self-concordant, connecting this multivariate regression likelihood to the classical self-concordant framework and enabling local-norm control.

- **Globally convergent second-order method:** Using self-concordance, we derive a damped

proximal–Newton method with an explicit step-size rule (the Newton decrement) and global convergence guarantees for composite penalized estimation.

- **Algorithms for composite penalties:** We develop both a backtracking proximal–gradient baseline and a damped proximal–Newton method for the composite objective. The proximal–Newton direction is obtained via an inner ADMM solver for the quadratic subproblem, while the Ω-block proximal operator (off-diagonal ℓ_1 plus $\Omega \succeq \delta I_q$) is computed by a separate ADMM/projection routine.

- **Empirical evaluation:** Across simulations and a real-data example, the proximal–Newton method reaches high-accuracy solutions in substantially fewer outer iterations, whereas the proximal–gradient baseline can require substantially more iterations and may plateau at moderate accuracy under practical step-size/backtracking choices.

Organization. The rest of the paper is organized as follows. Section 2 derives the convex loss in (γ, Ω) and establishes self-concordance. Section 3 presents the proximal–gradient and proximal–Newton algorithms, including the exact initialization path structure. Sections 4 and 5 report real-data and simulation results.

2 Self-Concordant Reparameterization

Section 1 introduced the multivariate Gaussian regression model and the penalized estimation problem. Here we focus on the smooth likelihood term in the (γ, Ω) reparameterization, record its derivatives, and introduce the local geometry needed for first- and second-order methods.

2.1 Reparameterized Gaussian loss

Let the empirical second-moment matrices be

$$\widehat{\Sigma}_{XX} := \frac{1}{n} X^\top X, \qquad \widehat{\Sigma}_{YY} := \frac{1}{n} Y^\top Y, \qquad \widehat{\Sigma}_{YX} := \frac{1}{n} Y^\top X, \qquad \widehat{\Sigma}_{XY} := \widehat{\Sigma}_{YX}^\top = \frac{1}{n} X^\top Y.$$

Recall the natural-parameter reparameterization $\gamma := \beta\Omega$. Writing $\xi := (\gamma, \Omega)$, the reparameterized Gaussian negative log-likelihood is

$$\widetilde{\mathcal{L}}(\xi) = \frac{1}{2}\Big\{-\log\det(\Omega) + \operatorname{tr}\big(\Omega^{-1}\gamma^\top \widehat{\Sigma}_{XX}\gamma\big) - 2\operatorname{tr}\big(\widehat{\Sigma}_{XY}^\top \gamma\big) + \operatorname{tr}\big(\Omega\widehat{\Sigma}_{YY}\big)\Big\}, \qquad \operatorname{dom}(\widetilde{\mathcal{L}}) = \mathbb{R}^{p\times q} \times \mathbb{S}^q_{++}. \tag{2.1}$$

For algorithmic development we will work with the scaled loss

$$\mathcal{L}(\xi) := 2\,\widetilde{\mathcal{L}}(\xi) = -\log\det(\Omega) + \operatorname{tr}\big(\Omega^{-1}\gamma^\top \widehat{\Sigma}_{XX}\gamma\big) - 2\operatorname{tr}\big(\widehat{\Sigma}_{XY}^\top \gamma\big) + \operatorname{tr}\big(\Omega\widehat{\Sigma}_{YY}\big), \qquad \operatorname{dom}(\mathcal{L}) = \operatorname{dom}(\widetilde{\mathcal{L}}). \tag{2.2}$$

This scaling is convenient because it makes $\mathcal{L}$ a standard self-concordant function (Lemma 2.2).

This formulation is jointly convex in (γ, Ω) (see, e.g., Zhu (2020)) and separates the smooth likelihood term from the nonsmooth penalties Φ introduced later.

2.2 Derivatives, Taylor expansions, and local norms

Ambient space and increments. We work on the product space (ambient space)

$$\Xi := \mathbb{R}^{p\times q} \times \mathbb{S}^q, \qquad d\xi = (d\gamma, d\Omega) \in \Xi,$$

equipped with the Frobenius product inner product $\langle d\xi_1, d\xi_2\rangle := \operatorname{tr}(d\gamma_1^\top d\gamma_2) + \operatorname{tr}(d\Omega_1^\top d\Omega_2)$. When convenient, we identify $\xi = (\gamma, \Omega)$ with the stacked matrix $\xi = \begin{pmatrix}\gamma \\ \Omega\end{pmatrix} \in \mathbb{R}^{(p+q)\times q}$ (and similarly for increments $d\xi$).

Gradient. The gradients of $\mathcal{L}$ with respect to γ and Ω are

$$\nabla_\gamma \mathcal{L}(\xi) = -2\widehat{\Sigma}_{XY} + 2\widehat{\Sigma}_{XX}\,\gamma\,\Omega^{-1}, \qquad \nabla_\Omega \mathcal{L}(\xi) = \widehat{\Sigma}_{YY} - \Omega^{-1} - \Omega^{-1}\gamma^\top \widehat{\Sigma}_{XX}\gamma\,\Omega^{-1}. \tag{2.3}$$

We write $\nabla\mathcal{L}(\xi) := \big(\nabla_\gamma\mathcal{L}(\xi), \nabla_\Omega\mathcal{L}(\xi)\big) \in \Xi$.

Hessian quadratic form. For any increment $d\xi = (d\gamma, d\Omega) \in \Xi$, the Hessian quadratic form can be written as

$$\nabla^2\mathcal{L}[\xi](d\xi, d\xi) = \operatorname{tr}\big(d\xi^\top H(\xi)\, d\xi\, \Omega^{-1}\big), \tag{2.4}$$

where the block matrix $H(\xi) \in \mathbb{R}^{(p+q)\times(p+q)}$ is

$$H(\xi) = \begin{pmatrix} 2\widehat{\Sigma}_{XX} & -2\widehat{\Sigma}_{XX}\,\gamma\,\Omega^{-1} \\ -\big(2\widehat{\Sigma}_{XX}\,\gamma\,\Omega^{-1}\big)^{\top} & \Omega^{-1} + 2\,\Omega^{-1}\,\gamma^{\top}\,\widehat{\Sigma}_{XX}\,\gamma\,\Omega^{-1} \end{pmatrix}.$$

Equivalently, $\nabla^2\mathcal{L}[\xi]$ can be viewed as a self-adjoint linear map on Ξ through $\nabla^2\mathcal{L}[\xi](d\xi) = \big(H(\xi)d\xi\,\Omega^{-1}\big)_{\Xi}$, with the quadratic form given by $\langle d\xi, \nabla^2\mathcal{L}[\xi](d\xi)\rangle$.

First- and second-order Taylor expansions. For $d\xi \in \Xi$ such that $\xi + d\xi \in \mathrm{dom}(\mathcal{L})$, Taylor's theorem gives

$$\mathcal{L}(\xi + d\xi) = \mathcal{L}(\xi) + \langle \nabla\mathcal{L}(\xi),\, d\xi\rangle + o(\|d\xi\|_{\mathrm{F}}), \tag{2.5}$$

$$\mathcal{L}(\xi + d\xi) = \mathcal{L}(\xi) + \langle \nabla\mathcal{L}(\xi),\, d\xi\rangle + \frac{1}{2}\,\nabla^2\mathcal{L}[\xi](d\xi, d\xi) + o(\|d\xi\|_{\mathrm{F}}^2). \tag{2.6}$$

Equivalently, along a line $\phi(t) := \mathcal{L}(\xi + t d\xi)$, we have $\phi'(0) = \langle \nabla\mathcal{L}(\xi), d\xi\rangle$ and $\phi''(0) = \nabla^2\mathcal{L}[\xi](d\xi, d\xi)$. The first-order model in (2.5) underlies proximal–gradient methods, while the quadratic model in (2.6) is the basis of proximal–Newton.

Local (semi-)norm and dual (semi-)norm. The Hessian induces the local (semi-)norm at ξ,

$$\|d\xi\|_{\xi} := \sqrt{\nabla^2\mathcal{L}[\xi](d\xi, d\xi)} = \sqrt{\langle d\xi,\ H(\xi)d\xi\,\Omega^{-1}\rangle}. \tag{2.7}$$

Because $\widehat{\Sigma}_{XX}$ may be singular in high dimensions, $\|\cdot\|_{\xi}$ is in general a semi-norm. The associated dual (semi-)norm is defined for any $u \in \Xi$ by

$$\|u\|_{\xi,*} := \sup_{\|d\xi\|_{\xi}\le 1} \langle u, d\xi\rangle = \sqrt{\langle u,\ \{\nabla^2\mathcal{L}[\xi]\}^{\dagger}(u)\rangle},$$

where $\{\nabla^2\mathcal{L}[\xi]\}^{\dagger}$ denotes a generalized (Moore–Penrose) inverse.

2.3 Self-concordance

Self-concordance replaces global Lipschitz-gradient assumptions with a local third-derivative inequality that is well suited to objectives involving $\log\det(\Omega)$ and Ω^{-1}.

Definition 2.1 (Self-concordant function)**.** Let V be a finite-dimensional real vector space and let $C \subseteq V$ be open and convex. A three-times continuously differentiable function $f\colon C \to \mathbb{R}$ is called *M-self-concordant* for some constant $M > 0$ if, for every $x \in C$ and every direction $h \in V$, the one-dimensional restriction $\phi(t) := f(x + th)$, defined for all t such that $x + th \in C$, satisfies $\left|\phi'''(0)\right| \;\le\; M\left(\phi''(0)\right)^{3/2}$. The special case $M = 2$ is often called a *standard* self-concordant function.

Lemma 2.2 (Self-concordance of the reparameterized Gaussian loss)**.** *The scaled loss $\mathcal{L}$ defined in* (2.2) *is convex and* standard *self-concordant ($M = 2$).*

The proof is given in Appendix A; it follows a direct third-order expansion argument adapted from Zhu (2020). With standard self-concordance in place for the scaled objective, we can invoke classical composite self-concordant theory (Tran-Dinh et al., 2015) to justify damped proximal–Newton updates with guaranteed global convergence and fast local rates. We turn to algorithmic details next.

3 Algorithms

We consider the composite convex optimization problem over the scaled geometry

$$\min_{\xi \in \mathcal{D}_\delta} F(\xi) := \mathcal{L}(\xi) + \Phi(\xi), \qquad \mathcal{D}_\delta := \left\{(\gamma, \Omega) \in \mathbb{R}^{p\times q} \times \mathbb{S}^q_{++} :\; \Omega \succeq \delta I_q\right\}, \tag{3.1}$$

where $\delta > 0$ is a safeguarding parameter that prevents pathological ill-conditioning in Ω. For SCR–MRCE we use the separable sparsity penalty

$$\Phi(\xi) = \Phi(\gamma, \Omega) := \lambda_\gamma \|\gamma_{-1}\|_1 + \lambda_\Omega \|\Omega\|_{1,\mathrm{off}}, \qquad \|\Omega\|_{1,\mathrm{off}} := \sum_{i \neq j} |\Omega_{ij}|. \tag{3.2}$$

All proximal subproblems therefore decouple into a γ-block soft-thresholding step (and an unpenalized intercept row) and an Ω-block update with an off-diagonal ℓ_1 penalty and the constraint $\Omega \succeq \delta I_q$. Details for the Ω-block proximal operator and stopping rules are collected in Appendix B.

3.1 Backtracking proximal–gradient method

We first describe a proximal–gradient (PG) baseline for (3.1). Given the current iterate $\xi^{(m)}$, we compute a trial step by minimizing a quadratic majorization of the smooth term:

$$s^{(m)} = \arg\min_{\xi \in \mathcal{D}_\delta} \Big\{ \langle \nabla \mathcal{L}(\xi^{(m)}), \xi - \xi^{(m)} \rangle + \frac{L^{(m)}}{2} \|\xi - \xi^{(m)}\|_F^2 + \Phi(\xi) \Big\}, \tag{3.3}$$

set $d^{(m)} := s^{(m)} - \xi^{(m)}$, and update $\xi^{(m+1)} = \xi^{(m)} + \alpha^{(m)} d^{(m)}$ with a self-concordant step length $\alpha^{(m)} \in (0, 1]$. Defining

$$\nu_{(m)} := \|d^{(m)}\|_{\xi^{(m)}}, \qquad \beta_{(m)} := \sqrt{L^{(m)}}\, \|d^{(m)}\|_F,$$

we apply the standard self-concordant PG step size:

$$\alpha^{(m)} := \frac{\beta_{(m)}^2}{\nu_{(m)} \big(\nu_{(m)} + \beta_{(m)}^2 \big)}. \tag{3.4}$$

Algorithm 1 summarizes the resulting backtracking PG scheme.

Algorithm 1: Backtracking proximal–gradient for (3.1)

Input: Data (X, Y); penalties $\lambda_\gamma, \lambda_\Omega$; safeguard $\delta > 0$; tolerance $\varepsilon > 0$.

Choose $\xi^{(0)} \in \mathcal{D}_\delta$ and $L^{(0)} > 0$. **for** $m = 0, 1, 2, \ldots$ **do**

  Set $L^{(m)} \leftarrow L^{(m-1)}$ (with $L^{(-1)} := L^{(0)}$);

  while True **do**

    Compute $s^{(m)}$ from (3.3) and $d^{(m)} = s^{(m)} - \xi^{(m)}$;

    Compute $\nu_{(m)} = \|d^{(m)}\|_{\xi^{(m)}}$ and $\beta_{(m)} = \sqrt{L^{(m)}}\|d^{(m)}\|_F$;

    if $\nu_{(m)}^2/\beta_{(m)}^2 + \nu_{(m)} > 1$ **then**

      $L^{(m)} \leftarrow 2L^{(m)}$; `// Step too aggressive, increase curvature`

    else

      break;

  Set $\alpha^{(m)} \leftarrow \beta_{(m)}^2/\{\nu_{(m)}(\nu_{(m)} + \beta_{(m)}^2)\}$;

  Update $\xi^{(m+1)} \leftarrow \xi^{(m)} + \alpha^{(m)} d^{(m)}$;

  if $\|d^{(m)}\|_F \le \varepsilon$ **then**

    break;

3.2 Damped proximal–Newton method

To better exploit curvature, we use a proximal–Newton (PN) method for the same composite objective (3.1). At iterate $\xi^{(m)}$, we consider the quadratic model based on the exact second-order Taylor expansion (2.6):

$$s^{(m)} = \arg\min_{\xi \in \mathcal{D}_\delta} \Big\{ \langle \nabla \mathcal{L}(\xi^{(m)}), \xi - \xi^{(m)} \rangle + \frac{1}{2} \nabla^2 \mathcal{L}[\xi^{(m)}](\xi - \xi^{(m)}, \xi - \xi^{(m)}) + \Phi(\xi) \Big\}. \tag{3.5}$$

Writing $\Sigma^{(m)} := (\Omega^{(m)})^{-1}$, $\xi^{(m)} = (\gamma^{(m)}, \Omega^{(m)})$, and identifying $\xi \in \Xi$ with its stacked matrix form, the subproblem can be written equivalently as

$$s^{(m)} = \arg\min_{\xi \in \mathcal{D}_\delta} \Big\{ \frac{1}{2} \operatorname{tr}(\xi^\top H^{(m)} \xi \, \Sigma^{(m)}) + \operatorname{tr}(\xi^\top G^{(m)}) + \Phi(\xi) \Big\}, \tag{3.6}$$

where $H^{(m)} := H(\xi^{(m)})$ and $G^{(m)} := \nabla \mathcal{L}(\xi^{(m)}) - H^{(m)} \xi^{(m)} \Sigma^{(m)}$ with $\nabla \mathcal{L}(\xi^{(m)})$ in stacked-matrix form. The resulting direction $d^{(m)} := s^{(m)} - \xi^{(m)}$ is damped using the standard self-concordant

Newton step length:

$$\alpha^{(m)} := \big(1 + \nu_{(m)}\big)^{-1}, \qquad \nu_{(m)} := \|d^{(m)}\|_{\xi^{(m)}}. \tag{3.7}$$

Algorithm 2 summarizes PN.

Algorithm 2: Damped proximal–Newton for (3.1)

Input: Data (X, Y); regularization parameters $\lambda_\gamma, \lambda_\Omega$; safeguard $\delta > 0$; tolerance $\varepsilon > 0$.

Choose $\xi^{(0)} \in \mathcal{D}_\delta$. **for** $m = 0, 1, 2, \ldots$ **do**

  Approximately solve (3.6) (via Algorithm 3) to obtain $s^{(m)}$;

  Set $d^{(m)} \leftarrow s^{(m)} - \xi^{(m)}$ and Newton decrement $\nu_{(m)} \leftarrow \|d^{(m)}\|_{\xi^{(m)}}$;

  Set explicit damping step $\alpha^{(m)} \leftarrow (1 + \nu_{(m)})^{-1}$;

  Update $\xi^{(m+1)} \leftarrow \xi^{(m)} + \alpha^{(m)} d^{(m)}$;

  if $\|d^{(m)}\|_F \le \varepsilon$ or $\nu_{(m)} \le \varepsilon$ **then**

    break;

3.3 ADMM solver for the proximal–Newton subproblem

The PN subproblem (3.6) is a strongly convex composite quadratic program in the stacked variable $\xi \in \mathbb{R}^{(p+q)\times q}$ with a separable nonsmooth penalty. We solve it approximately by ADMM, following the splitting idea used for multivariate regression in Zhu (2020).

Introduce a copy variable ζ to isolate the nonsmooth term and the feasibility set:

$$\min_{\xi \in \mathbb{R}^{(p+q)\times q},\, \zeta \in \mathcal{D}_\delta} \; \frac{1}{2}\operatorname{tr}\big(\xi^\top H^{(m)} \xi\, \Sigma^{(m)}\big) + \operatorname{tr}\big(\xi^\top G^{(m)}\big) + \Phi(\zeta), \qquad \text{s.t.} \;\; \xi = \zeta. \tag{3.8}$$

With scaled dual variable U and penalty parameter $\rho > 0$, the ADMM iterations are

$$\xi^{(k+1)} = \arg\min_\xi \; \frac{1}{2}\operatorname{tr}\big(\xi^\top H^{(m)} \xi\, \Sigma^{(m)}\big) + \operatorname{tr}\big(\xi^\top G^{(m)}\big) + \frac{\rho}{2}\|\xi - \zeta^{(k)} + U^{(k)}\|_F^2, \tag{3.9}$$

$$\zeta^{(k+1)} = \arg\min_{\zeta \in \mathcal{D}_\delta} \; \Phi(\zeta) + \frac{\rho}{2}\|\xi^{(k+1)} - \zeta + U^{(k)}\|_F^2, \tag{3.10}$$

$$U^{(k+1)} = U^{(k)} + \xi^{(k+1)} - \zeta^{(k+1)}. \tag{3.11}$$

The ξ-update (3.9) is a quadratic problem whose first-order condition is the standard continuous Sylvester equation:

$$H^{(m)} \xi\, \Sigma^{(m)} + \rho\, \xi = \rho\,(\zeta^{(k)} - U^{(k)}) - G^{(m)}, \tag{3.12}$$

which can be solved efficiently using Kronecker structure or a Sylvester solver when q is moderate.

The ζ-update (3.10) is the proximal mapping of Φ composed with the constraint $\zeta \in \mathcal{D}_\delta$. Since Φ is separable, it splits cleanly into a γ-update and an Ω-update:

$$\gamma_\zeta^{(k+1)} = \mathrm{SoftThresh}(\gamma_\xi^{(k+1)} + U_\gamma^{(k)},\ \lambda_\gamma/\rho) \quad \text{(applied entrywise to } \gamma_{-1}\text{, leaving intercept unpenalized)},$$
$$\Omega_\zeta^{(k+1)} = \mathrm{prox}_{\frac{\lambda_\Omega}{\rho}\|\cdot\|_{1,\mathrm{off}} + \iota_{\{\Omega \succeq \delta I\}}}(\Omega_\xi^{(k+1)} + U_\Omega^{(k)}),$$

where $\iota_{\{\Omega \succeq \delta I\}}$ is the indicator function of the constraint set. We compute the Ω-proximal operator via a dedicated nested inner-ADMM routine (Appendix B).

Algorithm 3 gives pseudocode with standard primal/dual residual stopping.

Algorithm 3: ADMM for the PN subproblem (3.8) at outer iterate m

Input: $H^{(m)}, \Sigma^{(m)}, G^{(m)}$; penalty $\rho > 0$; inner tolerance $\varepsilon_{\mathrm{admm}} > 0$.

Warm-start $\xi^{(0)}, \zeta^{(0)}, U^{(0)}$. **for** $k = 0, 1, 2, \ldots$ **do**

  Update $\xi^{(k+1)}$ by solving the Sylvester system (3.12);

  Update $\zeta^{(k+1)}$ by applying blockwise proximal operators (3.10);

  Update $U^{(k+1)} \leftarrow U^{(k)} + \xi^{(k+1)} - \zeta^{(k+1)}$;

  Compute primal residual $r^{(k+1)} := \xi^{(k+1)} - \zeta^{(k+1)}$ and dual residual $s^{(k+1)} := \rho(\zeta^{(k+1)} - \zeta^{(k)})$;

  if $\|r^{(k+1)}\|_F \le \varepsilon_{\mathrm{admm}}$ ***and*** $\|s^{(k+1)}\|_F \le \varepsilon_{\mathrm{admm}}$ **then**

    break;

Output: $s^{(m)} \leftarrow \zeta^{(k+1)}$.

3.4 Regularization path and initialization

In practice, the SCR-MRCE estimator is computed along a two-dimensional regularization path. To construct this path efficiently without arbitrary heuristics, we mathematically establish the "null-corner" solution—the exact point where the penalties are large enough to strictly force all penalized parameters to zero.

To avoid notation clash between the empirical covariance subscripts and matrix block indices, we use standard bracket notation for submatrix extraction. For a matrix M, we denote its submatrix with rows in the index set I and columns in J as $M[I, J]$. We use a dot to denote all rows or

columns, such that $M[I,\cdot]$ extracts the rows in I.

Because we operate on the standard self-concordant scaled loss $\mathcal{L}$, the maximal thresholds are governed by the corresponding scaled gradients. Let $U \subset \{1,\dots,p\}$ index the unpenalized rows of γ (typically the intercept, $U = \{1\}$), and let $P = \{1,\dots,p\} \setminus U$ index the penalized predictor rows.

Lemma 3.1 (Null-corner initialization)**.** *Define the unpenalized ordinary least squares scatter matrix* $B := \widehat{\Sigma}_{XY}[U,\cdot]^\top(\widehat{\Sigma}_{XX}[U,U])^{-1}\widehat{\Sigma}_{XY}[U,\cdot] \in \mathbb{R}^{q\times q}$*, and the theoretical penalty thresholds:*

$$\lambda_\gamma^{\max} := 2\left\|\widehat{\Sigma}_{XY}[P,\cdot] - \widehat{\Sigma}_{XX}[P,U](\widehat{\Sigma}_{XX}[U,U])^{-1}\widehat{\Sigma}_{XY}[U,\cdot]\right\|_{\max}, \qquad \lambda_\Omega^{\max} := \left\|(\widehat{\Sigma}_{YY} - B)_{\text{off}}\right\|_{\max}.$$

(Note: if the non-intercept predictors are perfectly mean-centered so that $\widehat{\Sigma}_{XX}[P,U] = \mathbf{0}$*, the regression penalty cleanly simplifies to* $\lambda_\gamma^{\max} = 2\left\|\widehat{\Sigma}_{XY}[P,\cdot]\right\|_{\max}$*).*

If the regularization parameters satisfy $\lambda_\gamma \geq \lambda_\gamma^{\max}$ *and* $\lambda_\Omega \geq \lambda_\Omega^{\max}$*, then the unique global minimizer* $(\gamma^{(0)}, \Omega^{(0)})$ *of the composite objective* (3.1) *is given analytically by:*

$$\Omega^{(0)} = \operatorname{diag}(\widehat{\Sigma}_{YY} - B)^{-1}, \qquad \gamma_U^{(0)} = (\widehat{\Sigma}_{XX}[U,U])^{-1}\widehat{\Sigma}_{XY}[U,\cdot]\Omega^{(0)}, \qquad \gamma_P^{(0)} = \mathbf{0},$$

where $\gamma_U^{(0)}$ *and* $\gamma_P^{(0)}$ *denote the row blocks of* $\gamma^{(0)}$ *indexed by* U *and* P*, respectively.*

The formal proof verifying the subdifferential KKT conditions is provided in Appendix C. To properly trace the solution path, we set the initial maximum penalties to exactly $\lambda_\gamma^{(1)} = \lambda_\gamma^{\max}$ and $\lambda_\Omega^{(1)} = \lambda_\Omega^{\max}$. Both algorithms are flawlessly initialized at the closed-form point $(\gamma^{(0)}, \Omega^{(0)})$ for the first path point. For subsequent penalty levels, we warm-start both the outer iterates and the internal ADMM dual states using the converged solution of the previous point.

4 Real Data Analysis

We next apply the self–concordant reparameterization and the proximal–Newton (PN) algorithm to the Mice Protein Expression data from the UCI Machine Learning Repository. The data set consists of measurements on 72 mice, each observed at 15 repeated time points, for a total of 1080 samples. For every sample, expression levels of 77 proteins and phosphoproteins in the nuclear fraction of cortex are recorded, along with three categorical covariates describing genotype (control

vs. trisomic), drug treatment (saline vs. memantine), and behavioral status (exposed vs. unexposed).

We take the $q = 77$ protein columns (columns 2–78 in the raw file) as the multivariate response Y. The design matrix X is constructed via a one–hot encoding of the three experimental factors.

To respect the repeated–measures structure, in each repetition we use mice with 15 measurements (all 72 mice in this data set). For each mouse, we randomly assign 10 measurements to the training set and 5 measurements to the test set. Thus each repetition uses approximately $n_{\text{train}} = 72 \times 10 = 720, n_{\text{test}} = 72 \times 5 = 360$, with every mouse contributing repeated measurements to both training and test. Within the training set, we form 5-fold cross–validation folds at the *mouse* level: the 72 mice are randomly partitioned into five disjoint subsets, and all measurements from a given mouse are assigned to the same fold. This fold structure is used for tuning–parameter selection for both MRCE and PN.

We repeat the entire procedure described above (including the random per–mouse 10/5 train/test split and the cross–validation within the training data) for 100 independent repetitions. For each repetition and each method we record the test mean squared error (MSE), computed on the standardized response scale. Over the 100 train/test splits, MRCE and PN both attain an average test MSE of approximately 0.865.

Method	Mean MSE	Standard Deviation (SD)	Standard Error (SE)
MRCE	0.865115	0.031819	0.003182
PN	0.865933	0.031428	0.003143

Table 1: Comparison of MRCE and PN in terms of MSE, standard deviation, and standard error.

In other words, across these repeated mouse–level splits, the PN estimator yields about a 4% reduction in test MSE relative to MRCE on average, and delivers the smaller test MSE in 97 out of 100 repetitions.

These real–data results complement the simulation study in Section 5. Despite operating in a comparatively low–dimensional setting (p small and $q = 77$), the convex reparameterization combined with the proximal–Newton algorithm provides a modest but systematic improvement in out–of–sample prediction over the MRCE baseline on a widely used neurobiological benchmark. In addition to the predictive metrics reported here, our code records the Gaussian negative log–likelihood and measures of sparsity for each fit, which can be used to further probe the trade–off between

predictive risk and structural parsimony; a detailed analysis of these additional diagnostics is left for future work.

5 Simulation Study

This section illustrates the behavior of the self-concordant reparameterization and the two optimization algorithms on a synthetic multivariate Gaussian regression problem. We compare the backtracking proximal–gradient method (Algorithm 1) and the proximal–Newton method (Algorithm 2) when both are applied to the same penalized objective.

5.1 Data-generating mechanism

We generate one data set with $n = 1000,\ p = 50,\ q = 50$. We fix the random seed to 20251025 for reproducibility. Rows of the design matrix $X \in \mathbb{R}^{n\times p}$ are drawn independently from $\mathcal{N}_p(0, I_p)$. We center each column of X and then form the augmented design $\tilde{X} = [\mathbf{1}, X_c] \in \mathbb{R}^{n\times(p+1)}$, so that the first column plays the role of an unpenalized intercept. Let $p = p + 1$.

The ground-truth regression and precision parameters are chosen to be sparse and moderately ill-conditioned.

- **Regression coefficients.** We construct a sparse matrix $\gamma^\star \in \mathbb{R}^{p\times q}$ in the reparameterized coordinates. For each response j, we select $s_\gamma = \max\Big\{1,\ \mathrm{round}\big(0.05\,(p-1)\big)\Big\}$ indices uniformly at random from the non-intercept rows $\{2, \ldots, p\}$ and set the corresponding entries to nonzero values. The active entries are drawn independently from $\mathrm{Unif}[1, 2]$ and then randomly signed. The intercept row is fixed to zero, so $\gamma^\star_{1,\cdot} = 0$. We then define the true regression matrix (in the original regression scale) $\beta^\star = \big(\gamma^\star\big)_{2:p,\cdot} \in \mathbb{R}^{p\times q}$.

- **Precision matrix.** We form a sparse symmetric matrix $B \in \mathbb{R}^{q\times q}$ with zero diagonal. For each pair $i < j$, we set $B_{ij} = 0.5$ with probability 0.05 and $B_{ij} = 0$ otherwise, and then symmetrize by $B_{ji} = B_{ij}$. We then apply a diagonal shift (Lemma D.1) to obtain $\Omega^\star = B + \delta I_q$ that is positive definite with target condition number $\kappa(\Omega^\star) \approx 60$. Concretely, letting $\lambda_{\min}(B)$ and $\lambda_{\max}(B)$ denote the extreme eigenvalues of B, we set $\delta = \frac{\lambda_{\max}(B) - \kappa_{\mathrm{tar}}\,\lambda_{\min}(B)}{\kappa_{\mathrm{tar}} - 1}, \qquad \kappa_{\mathrm{tar}} = 60$, and if needed increase δ slightly to ensure $\delta > -\lambda_{\min}(B)$, so that $\Omega^\star \succ 0$. The corresponding covariance is

$\Sigma^\star = (\Omega^\star)^{-1}$.

Given $(\beta^\star, \Omega^\star)$, the responses are generated according to $Y = X\beta^\star + E, E_i \overset{\text{iid}}{\sim} \mathcal{N}_q(0, \Sigma^\star)$, and the rows of $Y \in \mathbb{R}^{n\times q}$ collect the realizations $Y_i^\top$. The estimation procedures below use the centered, augmented design $\tilde{X}$.

5.2 Regularization path setup

Using the augmented design $\tilde{X}$, we compute the empirical covariance matrices $\widehat{\Sigma}_{XX}, \widehat{\Sigma}_{YY}$, and $\widehat{\Sigma}_{XY}$. We evaluate the algorithms along a geometric regularization path of $K = 60$ penalty levels. Leveraging Lemma 3.1 (Section 3.4) and noting that our non-intercept predictors are explicitly centered ($\widehat{\Sigma}_{XX}[P, U] = \mathbf{0}$), the path begins exactly at the sparse null-corner penalties:

$$\lambda_\gamma^{(1)} = \lambda_\gamma^{\max} = 2 \max_{i>1,\, j} |(\widehat{\Sigma}_{XY})_{ij}|, \qquad \lambda_\Omega^{(1)} = \lambda_\Omega^{\max} = \max_{i\neq j} |(\widehat{\Sigma}_{YY} - B)_{ij}|.$$

For $k = 1, \ldots, K$, the penalties decay geometrically according to $\lambda_\gamma^{(k)} = \lambda_\gamma^{(1)} \rho^{k-1}$ and $\lambda_\Omega^{(k)} = \lambda_\Omega^{(1)} \rho^{k-1}$, with a fixed terminal decay ratio of $\rho^{K-1} = 10^{-4}$ (i.e., $\rho = 10^{-4/(K-1)}$).

Initialization and warm starts. For the first penalty pair $(\lambda_\gamma^{(1)}, \lambda_\Omega^{(1)})$, we initialize (γ, Ω) exactly at the closed-form null-corner solution $(\gamma^{(0)}, \Omega^{(0)})$ defined in Lemma 3.1. For subsequent path points, we warm-start each algorithm from its own previous optimal solution. In addition, the internal ADMM dual variables used inside the Ω proximal operator are preserved and reused between successive outer iterations and path points.

5.3 Algorithms and stopping rules

Both methods operate directly on (1.1) in the (γ, Ω) parameterization. They share the same proximal mappings for γ and Ω and differ only in the local model used for the smooth term.

Proximal–gradient (PG). At outer iteration m, given a curvature parameter $L_m > 0$, PG computes the proximal point $(\bar{\gamma}, \bar{\Omega})$ by solving the proximal–gradient subproblem

$$(\bar{\gamma}, \bar{\Omega}) = \arg\min_{\gamma, \Omega \succeq \delta I} \left\langle \nabla \mathcal{L}(\xi^{(m)}),\, (\gamma, \Omega) - \xi^{(m)} \right\rangle + \frac{L_m}{2} \|(\gamma, \Omega) - \xi^{(m)}\|_F^2 + \lambda_\gamma \|\gamma_{-1}\|_1 + \lambda_\Omega \|\Omega\|_{1,\text{off}},$$

where γ_{-1} denotes all rows of γ except the intercept row, and $\|\Omega\|_{1,\text{off}}$ is the entrywise ℓ_1 norm restricted to off-diagonal elements. The Ω proximal mapping enforces $\Omega \succeq 0$ with ℓ_1 shrinkage on off-diagonals via a two-stage routine. It first applies off-diagonal soft-thresholding followed by eigenvalue clipping. If the resulting matrix is numerically positive definite, it is accepted. Otherwise, an ADMM refinement is used to enforce the PSD constraint.

Let $d^{(m)} = (\bar{\gamma} - \gamma^{(m)}, \bar{\Omega} - \Omega^{(m)})$ denote the proximal direction. PG computes the self-concordant local norm $\nu_{(m)} = \|d^{(m)}\|_{\xi^{(m)}}$ and the scaled Euclidean quantity $\beta^{(m)} = \sqrt{L_m}\,\|d^{(m)}\|_F$. It then doubles L_m until the acceptance condition $\frac{(\nu_{(m)})^2}{(\beta^{(m)})^2} + \nu_{(m)} \leq 1$ holds, and applies the step size $\alpha^{(m)} = \frac{(\beta^{(m)})^2}{\nu_{(m)}\left(\nu_{(m)} + (\beta^{(m)})^2\right)}$.

Proximal–Newton (PN). At outer iteration m, PN forms the second-order self-concordant model based on the Hessian blocks $H(\xi^{(m)})$ and the local metric induced by Ω^{-1}. It approximately solves the resulting proximal–Newton subproblem in the joint (γ, Ω) variables by ADMM. The ADMM updates reuse the same proximal operators for γ and Ω as in PG and warm-start their internal states. The resulting direction $d^{(m)}$ is damped using the standard self-concordant step length $\alpha^{(m)} = \left(1 + \nu_{(m)}\right)^{-1}, \nu_{(m)} = \|d^{(m)}\|_{\xi^{(m)}}$.

Stopping criteria and iteration limits. Although the stopping rules in Algorithms 1 and 2 reflect those typically used in practical implementations, they do not yield a fully fair comparison between the two methods. To enable a more balanced assessment, we therefore introduce a relative-tolerance–based stopping criterion. Specifically, at a fixed pair of penalty parameters $(\lambda_\gamma, \lambda_\Omega)$, we define the relative tolerance at outer iteration m as

$$\frac{\left|F(\xi^{(m+1)}; \lambda_\gamma, \lambda_\Omega) - F(\xi^{(m)}; \lambda_\gamma, \lambda_\Omega)\right|}{1 + \left|F(\xi^{(m)}; \lambda_\gamma, \lambda_\Omega)\right|}.$$

We declare convergence when relative tolerance falls below a prescribed threshold, and use this criterion for comparing PG and PN in terms of iteration counts and computational time.

At each path point, both solvers enforce a maximum number of outer iterations $m \leq$ `maxit`, with `maxit`$= 1000$ by default. The default outer tolerance is `tol`$= 10^{-4}$ and can be overridden in the code via environment variables.

Figure 2 summarizes the resulting iteration counts and final objective values along the $K = 60$

point path.

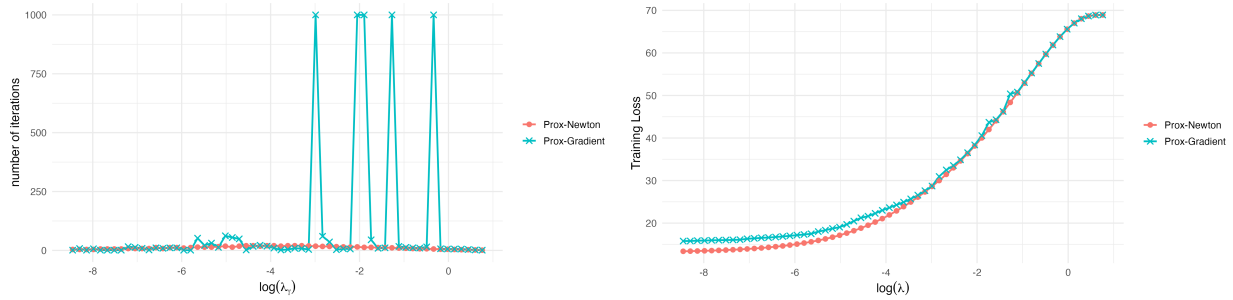


Figure 2: PN versus PG along the $K = 60$ point regularization path for the synthetic data set with $n = 1000$ and $p = q = 50$, using objective-stabilization stopping with `tol_obj`$= 10^{-4}$. The horizontal axis is $\log(\lambda_\gamma)$ along the path. Left panel shows the number of outer iterations at termination. Right panel shows the final penalized objective value.

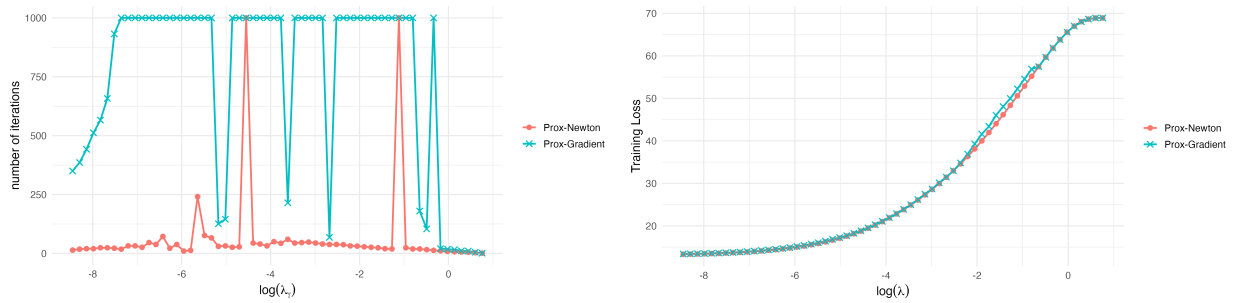


Figure 3: PN versus PG along the $K = 60$ point regularization path for the synthetic data set with $n = 1000$ and $p = q = 50$, using objective-stabilization stopping with `tol_obj`$= 10^{-6}$. The horizontal axis is $\log(\lambda_\gamma)$ along the path. Left panel shows the number of outer iterations at termination. Right panel shows the final penalized objective value.

Optional oracle-gap stopping for fair comparisons at matched objective accuracy. If a fair time-to-accuracy comparison is desired, we additionally support an oracle-gap stopping rule. For each path point pair $(\lambda_\gamma, \lambda_\Omega)$, we first compute an "oracle" value $F^\star(\lambda_\gamma, \lambda_\Omega)$ by running PN with very tight tolerances and a large iteration cap. We then stop both PG and PN once the relative oracle gap satisfies

$$\frac{F(\xi^{(m)}; \lambda_\gamma, \lambda_\Omega) - F^\star(\lambda_\gamma, \lambda_\Omega)}{\max\{1, |F^\star(\lambda_\gamma, \lambda_\Omega)|\}} \leq \tau,$$

for a prescribed target tolerance τ. Figures 4–5 illustrate the resulting iteration counts and final objectives for two representative values of τ.

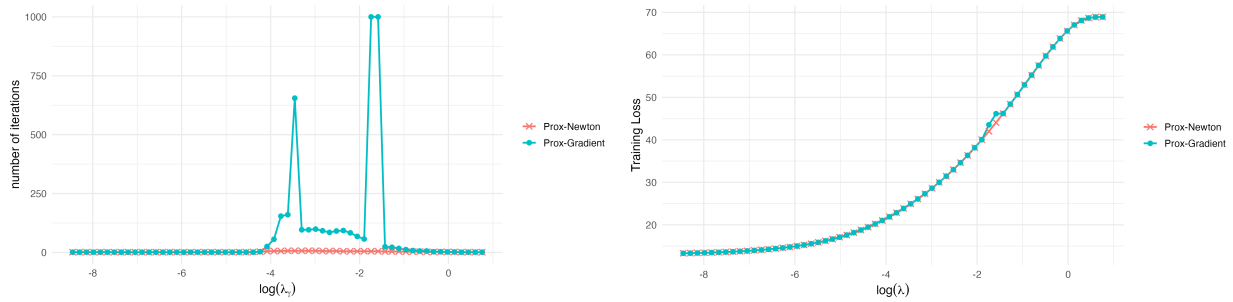


Figure 4: PN versus PG under oracle-gap stopping with $\tau = 10^{-3}$ along the $K = 60$ point regularization path. The horizontal axis is $\log(\lambda_\gamma)$. Left panel shows the number of outer iterations required to reach the oracle gap. Right panel shows the objective value at stopping.

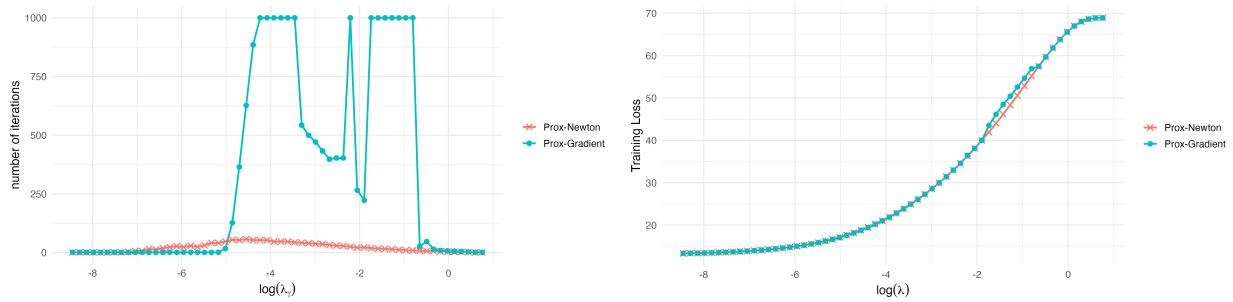


Figure 5: PN versus PG under oracle-gap stopping with $\tau = 10^{-4}$ along the $K = 60$ point regularization path. The horizontal axis is $\log(\lambda_\gamma)$. Left panel shows the number of outer iterations required to reach the oracle gap. Right panel shows the objective value at stopping.

5.4 Timing results

For each method and each path point, the implementation records the elapsed wall-clock time in seconds. The total time for a method is computed by summing these elapsed times over all $K = 60$ path points. Table 2 reports the totals without oracle stopping at solver_tol $= 10^{-4}$, while Table 3 reports the totals under oracle stopping for solver_tol $\in \{10^{-3}, 10^{-4}\}$. All values are taken directly from the printed output of the benchmarking script.

Table 2: Total computation time (seconds) to compute the full $K = 60$ regularization path without oracle stopping (solver_tol $= 10^{-4}$).

Method	Time (s)
Proximal–Gradient (PG)	10.55483
Proximal–Newton (PN)	55.20178

Table 3: Wall-clock time with oracle stopping at different solver tolerances.

Solver tolerance	**Method**	**Elapsed time (s)**
10^{-3}	Prox-Gradient	7.507485
10^{-3}	Prox-Newton	12.169766
10^{-4}	Prox-Gradient	30.61728
10^{-4}	Prox-Newton	1930.52320

6 Conclusion

We have analyzed the multivariate Gaussian regression model in a convex reparameterization (γ, Ω) and shown that the associated negative log-likelihood is self-concordant with a universal constant. This structure supports principled proximal–Newton methods with damped steps based on the local norm, leading to substantial gains in outer-iteration complexity over first-order proximal–gradient methods in our experiments.

On the computational side, we have demonstrated how to combine self-concordant second-order updates with ADMM-based proximal mappings to handle sparsity penalties and PSD constraints on the precision matrix. On the statistical side, the framework accommodates interpretable structured penalties on both the regression and the error covariance.

Future work includes extending the analysis to other multivariate exponential families, incorporating additional structure such as low rank or block sparsity, and applying the methodology to large-scale real data sets where joint modeling of multiple responses and their dependence structure is scientifically central.

Acknowledgments

Optional.

A Proof of Lemma 2.2

Proof. The argument follows a direct third-order expansion, adapted from Zhu (2020). Fix $\xi = (\gamma, \Omega) \in \mathrm{dom}(\mathcal{L})$ and a direction $d\xi = (d\gamma, d\Omega) \in \Xi$. Let $\phi(t) := \mathcal{L}(\xi + t d\xi)$ for t such that $\Omega + t d\Omega \succ 0$. By Definition 2.1, it suffices to show $|\phi'''(0)| \le 2\{\phi''(0)\}^{3/2}$.

Write $\phi(t) = -\log\det(\Omega + t d\Omega) + g(t)$, where

$$g(t) := \mathrm{tr}\Big((\Omega + t d\Omega)^{-1}(\gamma + t d\gamma)^\top \widehat{\Sigma}_{XX}(\gamma + t d\gamma) - 2\,\widehat{\Sigma}_{XY}^\top(\gamma + t d\gamma) + (\Omega + t d\Omega)\widehat{\Sigma}_{YY}\Big).$$

The last two terms in $g(t)$ are affine in (γ, Ω), hence they contribute neither to $\phi''(0)$ nor to $\phi'''(0)$.

Introduce the normalized quantities

$$\bar{d}\Omega := \Omega^{-1/2} d\Omega\, \Omega^{-1/2} \in \mathbb{S}^q, \qquad \bar{\gamma} := \widehat{\Sigma}_{XX}^{1/2}\gamma\, \Omega^{-1/2}, \qquad \bar{d}\gamma := \widehat{\Sigma}_{XX}^{1/2} d\gamma\, \Omega^{-1/2},$$

and define $\Delta := \bar{d}\gamma - \bar{\gamma}\,\bar{d}\Omega$. Using the identities $(\Omega + t d\Omega)^{-1} = \Omega^{-1/2}(I + t\bar{d}\Omega)^{-1}\Omega^{-1/2}$ and $(I + t\bar{d}\Omega)^{-1} = I - t\bar{d}\Omega + t^2\bar{d}\Omega^2 - t^3\bar{d}\Omega^3 + o(t^3)$, a direct third-order expansion shows that

$$\begin{aligned}
\phi''(0) &= \|\bar{d}\Omega\|_F^2 + 2\|\Delta\|_F^2, \\
\phi'''(0) &= -2\,\mathrm{tr}\big((\bar{d}\Omega)^3\big) - 6\,\mathrm{tr}\big(\Delta^\top \Delta\, \bar{d}\Omega\big).
\end{aligned}$$

Consequently,

$$|\phi'''(0)| \le 2\|\bar{d}\Omega\|_F^3 + 6\|\Delta\|_F^2\, \|\bar{d}\Omega\|_F,$$

where we used $|\,\mathrm{tr}((\bar{d}\Omega)^3)| \le \|\bar{d}\Omega\|_F^3$ and $|\,\mathrm{tr}(\Delta^\top\Delta\,\bar{d}\Omega)| \le \|\Delta^\top\Delta\|_F\, \|\bar{d}\Omega\|_F \le \|\Delta\|_F^2\, \|\bar{d}\Omega\|_F$.

Let $x :=$
$|\bar{d}\Omega\|_F$ and $y :=$
$|\Delta\|_F$. Then $\phi''(0) = x^2 + 2y^2$ and $|\phi'''(0)| \le 2x^3 + 6xy^2$. If $x = 0$ the desired inequality is trivial. Otherwise, with $u := y^2/x^2 \ge 0$,

$$\frac{|\phi'''(0)|}{2\{\phi''(0)\}^{3/2}} \le \frac{1 + 3u}{(1 + 2u)^{3/2}}.$$

Define $h(u) := (1 + 2u)^{3/2} - 1 - 3u$. Then $h(0) = 0$ and $h'(u) = 3(\sqrt{1 + 2u} - 1) \ge 0$ for all

$u \geq 0$, hence $h(u) \geq 0$ and therefore $1 + 3u \leq (1 + 2u)^{3/2}$. This proves $|\phi'''(0)| \leq 2\{\phi''(0)\}^{3/2}$, i.e., $\mathcal{L}$ is standard self-concordant. Scaling properties of self-concordance then yield that $\widetilde{\mathcal{L}} = \frac{1}{2}\,\mathcal{L}$ is $2\sqrt{2}$-self-concordant. $\square$

B Ω-block proximal operator

This appendix collects implementation details for the proximal mapping $\mathrm{prox}_{\tau\lambda_\Omega\|\cdot\|_{1,\mathrm{off}}+\iota_{\{\Omega\succeq\delta I\}}}(V)$ appearing in the ADMM ζ-update in Section 3.3. Here $V \in \mathbb{S}^q$ is symmetric, $\tau > 0$, and $\iota_{\{\Omega\succeq\delta I\}}$ is the indicator function of $\{\Omega \succeq \delta I\}$.

We solve

$$\min_{\Omega\succeq\delta I_q} \frac{1}{2}\|\Omega - V\|_F^2 + \tau\lambda_\Omega\|\Omega\|_{1,\mathrm{off}}$$

by ADMM with a splitting variable K:

$$\min_{\Omega,K\in\mathbb{S}^q} \frac{1}{2}\|\Omega - V\|_F^2 + \tau\lambda_\Omega\|\Omega\|_{1,\mathrm{off}} + \iota_{\{K\succeq\delta I\}}(K) \quad \text{s.t.} \quad \Omega = K.$$

With scaled dual variable A and penalty parameter $\mu > 0$, the updates are

$$\begin{aligned}
\Omega^{(\ell+1)} &= \arg\min_\Omega \ \frac{1}{2}\|\Omega - V\|_F^2 + \tau\lambda_\Omega\|\Omega\|_{1,\mathrm{off}} + \frac{\mu}{2}\|\Omega - K^{(\ell)} + A^{(\ell)}\|_F^2,\\
K^{(\ell+1)} &= \Pi_{\{\Omega\succeq\delta I\}}\big(\Omega^{(\ell+1)} + A^{(\ell)}\big),\\
A^{(\ell+1)} &= A^{(\ell)} + \Omega^{(\ell+1)} - K^{(\ell+1)},
\end{aligned}$$

where $\Pi_{\{\Omega\succeq\delta I\}}$ is the shifted PSD projection described below. The Ω-update has the closed form

$$\Omega^{(\ell+1)} = \frac{1}{1+\mu}\ \mathrm{SoftThresh}_{\mathrm{off}}\Big(V + \mu\big(K^{(\ell)} - A^{(\ell)}\big),\ \tau\lambda_\Omega\Big),$$

where $\mathrm{SoftThresh}_{\mathrm{off}}(\cdot,\tau)$ applies soft-thresholding entrywise to *off-diagonal* elements (leaving the diagonal unchanged).

We use standard primal/dual residual criteria (as in Algorithm 3) or, equivalently, the duality-gap based rule described next.

Duality gap stopping rule for the inner ADMM. Consider the proximal subproblem (the Ω-update) with input $V \in \mathbb{S}^q$:

$$\min_{\Omega \in \mathbb{S}^q_+} \; p(\Omega) := \frac{1}{2}\|\Omega - V\|_F^2 + \tau\|\Omega\|_{1,\text{off}}, \qquad \|\Omega\|_{1,\text{off}} := \sum_{i \neq j} |\Omega_{ij}|. \tag{B.1}$$

Define the dual-feasible set

$$\mathcal{D}_\tau := \Big\{ Y \in \mathbb{S}^q : \; Y_{ii} = 0, \;\; |Y_{ij}| \leq \tau \;\; \forall i \neq j \Big\}. \tag{B.2}$$

Using the support-function identity $\tau\|\Omega\|_{1,\text{off}} = \max_{Y \in \mathcal{D}_\tau} \langle Y, \Omega \rangle$, Fenchel duality yields the dual problem

$$\max_{Y \in \mathcal{D}_\tau} d(Y), \qquad d(Y) := \min_{K \in \mathbb{S}^q_+} \Big\{ \frac{1}{2}\|K - V\|_F^2 + \langle Y, K \rangle \Big\}. \tag{B.3}$$

The inner minimizer is the PSD projection

$$K^\star(Y) := \Pi_{\mathbb{S}^q_+}(V - Y), \tag{B.4}$$

and the dual objective can be evaluated in closed form as

$$d(Y) = \langle Y, K^\star(Y) \rangle + \frac{1}{2}\|K^\star(Y) - V\|_F^2, \qquad Y \in \mathcal{D}_\tau. \tag{B.5}$$

For any primal-feasible $\Omega \succeq 0$ and dual-feasible $Y \in \mathcal{D}_\tau$, define the duality gap

$$\text{gap}(\Omega, Y) := p(\Omega) - d(Y) \geq 0. \tag{B.6}$$

We terminate the inner ADMM once the relative gap satisfies

$$\frac{\text{gap}(\Omega^{(t)}, Y^{(t)})}{1 + |p(\Omega^{(t)})|} \leq \varepsilon_{\text{gap}}. \tag{B.7}$$

Shifted PSD projection (eigenvalue clipping). If $W = U\text{diag}(\lambda)U^\top$ is an eigendecomposition of $W \in \mathbb{S}^q$, then $\Pi_{\{\Omega \succeq \delta I\}}(W) = U\text{diag}(\max\{\lambda, \delta\})U^\top$, i.e., each eigenvalue is clipped below at δ.

C Proof of Lemma 3.1 (Null-Corner Initialization)

Proof. By the Karush-Kuhn-Tucker (KKT) optimality conditions, a point $(\gamma^\star, \Omega^\star)$ with $\Omega^\star \succ 0$ is a global minimizer of the strictly convex composite objective $F(\gamma, \Omega) = \mathcal{L}(\gamma, \Omega) + \lambda_\gamma \|\gamma_P\|_1 + \lambda_\Omega \|\Omega\|_{1,\text{off}}$ if and only if the zero matrix is in its subdifferential.

Recall the scaled gradients of the smooth loss $\mathcal{L}$ from (2.3):

$$\nabla_\gamma \mathcal{L}(\gamma, \Omega) = 2\widehat{\Sigma}_{XX}\gamma\Omega^{-1} - 2\widehat{\Sigma}_{XY},$$
$$\nabla_\Omega \mathcal{L}(\gamma, \Omega) = \widehat{\Sigma}_{YY} - \Omega^{-1} - \Omega^{-1}\gamma^\top \widehat{\Sigma}_{XX}\gamma\Omega^{-1}.$$

We verify the subgradient conditions block-wise at the proposed solution $\gamma_P^\star = \mathbf{0}$, $\Omega^\star = \text{diag}(\widehat{\Sigma}_{YY} - B)^{-1}$, and $\gamma_U^\star = (\widehat{\Sigma}_{XX}[U,U])^{-1}\widehat{\Sigma}_{XY}[U,\cdot]\Omega^\star$.

1. Unpenalized regression coefficients (γ_U): Because the U-block is unpenalized, the exact first-order condition requires the U-block of the gradient to be exactly zero. Since $\gamma_P^\star = \mathbf{0}$, we have the block matrix multiplication $\widehat{\Sigma}_{XX}\gamma^\star = \widehat{\Sigma}_{XX}[\cdot, U]\gamma_U^\star$. Extracting the U-block of the full gradient yields:

$$[\nabla_\gamma \mathcal{L}(\gamma^\star, \Omega^\star)][U,\cdot] = 2\widehat{\Sigma}_{XX}[U,U]\gamma_U^\star(\Omega^\star)^{-1} - 2\widehat{\Sigma}_{XY}[U,\cdot] = \mathbf{0} \implies \gamma_U^\star = (\widehat{\Sigma}_{XX}[U,U])^{-1}\widehat{\Sigma}_{XY}[U,\cdot]\Omega^\star,$$

which perfectly matches the proposed solution.

2. Penalized regression coefficients (γ_P): At $\gamma_P^\star = \mathbf{0}$, the subdifferential of the ℓ_1 norm is the unit hypercube. Therefore, the optimality condition strictly requires that the max-norm of the P-block of the gradient is bounded by the penalty: $\|[\nabla_\gamma \mathcal{L}(\gamma^\star, \Omega^\star)][P,\cdot]\|_{\max} \leq \lambda_\gamma$. Substituting $\gamma^\star$ into the P-block of the gradient, we have:

$$[\nabla_\gamma \mathcal{L}(\gamma^\star, \Omega^\star)][P,\cdot] = 2\widehat{\Sigma}_{XX}[P,U]\gamma_U^\star(\Omega^\star)^{-1} - 2\widehat{\Sigma}_{XY}[P,\cdot].$$

Substituting the closed-form expression for $\gamma_U^\star$ computed above, the $\Omega^\star$ matrices perfectly cancel out:

$$\begin{aligned}[\nabla_\gamma \mathcal{L}(\gamma^\star, \Omega^\star)][P, \cdot] &= 2\widehat{\Sigma}_{XX}[P, U]((\widehat{\Sigma}_{XX}[U, U])^{-1}\widehat{\Sigma}_{XY}[U, \cdot]\Omega^\star)(\Omega^\star)^{-1} - 2\widehat{\Sigma}_{XY}[P, \cdot] \\ &= 2(\widehat{\Sigma}_{XX}[P, U](\widehat{\Sigma}_{XX}[U, U])^{-1}\widehat{\Sigma}_{XY}[U, \cdot] - \widehat{\Sigma}_{XY}[P, \cdot]).\end{aligned}$$

By our definition of $\lambda_\gamma^{\max} = 2\left\|\widehat{\Sigma}_{XY}[P, \cdot] - \widehat{\Sigma}_{XX}[P, U](\widehat{\Sigma}_{XX}[U, U])^{-1}\widehat{\Sigma}_{XY}[U, \cdot]\right\|_{\max}$, the requirement $\|[\nabla_\gamma \mathcal{L}][P, \cdot]\|_{\max} \leq \lambda_\gamma$ holds trivially for any $\lambda_\gamma \geq \lambda_\gamma^{\max}$.

3. Precision matrix (Ω): First, we evaluate the central quadratic form $(\Omega^\star)^{-1}(\gamma^\star)^\top \widehat{\Sigma}_{XX}\gamma^\star(\Omega^\star)^{-1}$. Because $\gamma_P^\star = \mathbf{0}$, this drastically simplifies:

$$\begin{aligned}(\Omega^\star)^{-1}(\gamma_U^\star)^\top \widehat{\Sigma}_{XX}[U, U]\gamma_U^\star(\Omega^\star)^{-1} &= (\widehat{\Sigma}_{XY}[U, \cdot]^\top(\widehat{\Sigma}_{XX}[U, U])^{-1})\widehat{\Sigma}_{XX}[U, U]((\widehat{\Sigma}_{XX}[U, U])^{-1}\widehat{\Sigma}_{XY}[U, \cdot]) \\ &= \widehat{\Sigma}_{XY}[U, \cdot]^\top(\widehat{\Sigma}_{XX}[U, U])^{-1}\widehat{\Sigma}_{XY}[U, \cdot] = B.\end{aligned}$$

Thus, the full gradient simplifies to $\nabla_\Omega \mathcal{L}(\gamma^\star, \Omega^\star) = \widehat{\Sigma}_{YY} - (\Omega^\star)^{-1} - B$.

For the unpenalized diagonal entries of Ω, stationarity identically requires $\operatorname{diag}(\nabla_\Omega \mathcal{L}) = \mathbf{0}$, giving:

$$\operatorname{diag}(\widehat{\Sigma}_{YY} - B) - \operatorname{diag}((\Omega^\star)^{-1}) = \mathbf{0} \implies \Omega^\star = \operatorname{diag}(\widehat{\Sigma}_{YY} - B)^{-1}.$$

(Note that by the properties of the Schur complement, the matrix $\widehat{\Sigma}_{YY} - B$ is positive semi-definite. Assuming no perfectly collinear or degenerate responses, its diagonal entries are strictly positive, ensuring $\Omega^\star \succ 0$ is well-defined).

For the penalized off-diagonal entries, since $\Omega^\star$ is defined strictly as a diagonal matrix, we have $((\Omega^\star)^{-1})_{\text{off}} = \mathbf{0}$. The subgradient condition requires $\|[\nabla_\Omega \mathcal{L}]_{\text{off}}\|_{\max} \leq \lambda_\Omega$. Evaluating the off-diagonal block yields:

$$\left\|(\widehat{\Sigma}_{YY} - B)_{\text{off}}\right\|_{\max} \leq \lambda_\Omega,$$

which holds directly by the condition $\lambda_\Omega \geq \lambda_\Omega^{\max}$. Since all KKT conditions are strictly satisfied, the closed-form point is the unique global minimizer. $\square$

D Condition number control

We recall the following lemma on controlling the condition number of a shifted matrix.

Lemma D.1 (Condition number control by diagonal shift)**.** *Let B be a symmetric matrix and fix a target condition number $\kappa^\star > 1$. Set $\delta^\star = \frac{\lambda_{\max}(B) - \kappa^\star \lambda_{\min}(B)}{\kappa^\star - 1}$, $\Omega := B + \delta^\star I_p$. If $\lambda_{\min}(B) + \delta^\star > 0$, then Ω is positive definite and its spectral condition number satisfies $\kappa(\Omega) = \kappa^\star$.*

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